



CSCI 151 - Programming for Scientists and Engineers Ғалымдар мен инженерлерге арналған бағдарламалау Программирование для ученых и инженеров

General Course Information

School: School of Engineering and Digital Sciences

Course Type: Undergraduate Common Core Curriculum

Semester in which the course is offered: Fall & Spring

Course instructor(s): Ben Tyler

Course Description - English

The course introduces students to imperative programming and the development of basic algorithms using the C programming language. A detailed presentation of the language syntax will be given, along with the proper use of control structures, loops, functions, and recursion to solve problems. The creation and use of structures and arrays will also be covered, along with the development of libraries and multi-file programs.

Course Description - Kazakh

Курс студенттерді Си бағдарламалау тілін пайдалана отырып, базалық алгоритмдерді императивті бағдарламалаумен және әзірлемелермен таныстырады. Тіл синтаксисінің, сондай-ақ проблемаларды шешуге арналған басқару құрылымдарын, циклдерін, функцияларын және рекурсиясын дұрыс пайдаланудың егжей-тегжейлі сипаттамасы беріледі. Құрылымдар мен массивтерді жасау және пайдалану да, кітапханалар мен көп файлдық бағдарламаларды әзірлеу де қарастырылатын болады.

Course Description - Russian

Курс знакомит студентов с императивным программированием и разработкой базовых алгоритмов с использованием языка программирования Си. Будет дано подробное описание синтаксиса языка, а также правильное использование управляющих структур, циклов, функций и рекурсии для решения проблем. Также будет рассмотрено создание и использование структур и массивов, а также разработка библиотек и многофайловых программ.

Course Requisites

Conditions of Enrollment

Anti-requisites (if applicable): CSCI 111 (D and above)

Anti-requisites (if applicable): CSCI 115 (D and above)

Anti-requisites (if applicable): ENG 101 (B- and above)

Course Aims

1. To introduce students to imperative programming and proper program design using the C programming language.
2. To show students how basic computational problems can be solved through algorithms and structured types.
3. To introduce students to the basic concepts of how computer programs work, including the use of files and memory.

Course Learning Outcomes (CLOs)

Course Learning Outcomes	Graduate Attributes
1. Utilize modern software development tools and methods. Such skills will be useful in students' future computer science courses, as well as in their future careers.	GA1 - Possess an in-depth and sophisticated understanding of their domain of study. GA2 - Be intellectually agile, curious, creative and open-minded. GA3 - Be thoughtful decision makers who know how to involve others. GA4 - Be entrepreneurial, self-propelling and able to create new opportunities. GA8 - Be prepared to take a leading role in the development of their country.
2. Have proficiency in writing and explaining small to mid-sized structured programs which will incorporate fundamental language elements	GA1 - Possess an in-depth and sophisticated understanding of their domain of study. GA2 - Be intellectually agile, curious, creative and open-minded. GA3 - Be thoughtful decision makers who know how to involve others. GA4 - Be entrepreneurial, self-propelling and able to create new opportunities. GA8 - Be prepared to take a leading role in the development of their country.
3. Solve abstract computational problems through the use of programs.	GA1 - Possess an in-depth and sophisticated understanding of their domain of study. GA2 - Be intellectually agile, curious, creative and open-minded. GA3 - Be thoughtful decision makers who know how to involve others. GA4 - Be entrepreneurial, self-propelling and able to create new opportunities. GA8 - Be prepared to take a leading role in

	the development of their country.
4. Write their own recursive functions and analyze other's recursive code for correctness.	GA1 - Possess an in-depth and sophisticated understanding of their domain of study. GA2 - Be intellectually agile, curious, creative and open-minded. GA3 - Be thoughtful decision makers who know how to involve others. GA4 - Be entrepreneurial, self-propelling and able to create new opportunities. GA8 - Be prepared to take a leading role in the development of their country.
5. Identify and compare common algorithms used in searching and sorting, and learn how to implement them in C.	GA1 - Possess an in-depth and sophisticated understanding of their domain of study. GA2 - Be intellectually agile, curious, creative and open-minded. GA3 - Be thoughtful decision makers who know how to involve others. GA4 - Be entrepreneurial, self-propelling and able to create new opportunities. GA8 - Be prepared to take a leading role in the development of their country.
6. Use pointers and memory management in the C language and be able to implement and explain code using these features.	GA1 - Possess an in-depth and sophisticated understanding of their domain of study. GA2 - Be intellectually agile, curious, creative and open-minded. GA3 - Be thoughtful decision makers who know how to involve others. GA4 - Be entrepreneurial, self-propelling and able to create new opportunities. GA8 - Be prepared to take a leading role in the development of their country.

Mapping of Undergraduate Common Core Curriculum Courses

This course supports achievement of the following learning outcome(s):

Please indicate the learning outcome(s) supported by the course (tick all that apply). You may wish to have a look at the [Undergraduate Common Core Curriculum Framework](#) (UCCF).

Course Learning Outcomes	Program Learning Outcomes
1. Utilize modern software development tools and methods. Such skills will be useful in students' future computer science courses, as well as in their future careers.	1. Apply numerical and digital literacy skills
2. Have proficiency in writing and explaining small to mid-sized structured programs which will incorporate fundamental language elements	1. Apply numerical and digital literacy skills
3. Solve abstract computational problems	1. Apply numerical and digital literacy skills

through the use of programs.	
4. Write their own recursive functions and analyze other's recursive code for correctness.	1. Apply numerical and digital literacy skills
5. Identify and compare common algorithms used in searching and sorting, and learn how to implement them in C.	1. Apply numerical and digital literacy skills
6. Use pointers and memory management in the C language and be able to implement and explain code using these features.	1. Apply numerical and digital literacy skills

ECTS and Student Workload

Credits: 8

Contact Hours: 60

Self-Study Hours: 150

Total Hours: 210

Assessment Methods

Assessment Methods	Course Learning Outcomes (Reference #)
Lab Work and Programming Assignment	1, 2, 3, 4, 5, 6
Midterm exam(s)	2, 3, 4, 5, 6
Final Exam	2, 3, 4, 5, 6

Assessment Methods Weighting

Assessment Methods	Weighting
Lab Work and Programming Assignment	20%
Midterm exam(s)	40%
Final Exam	40%
Total weighting	100%

Course Outline

Session Number	Theme (Key Topic)	Course Learning Outcomes (Reference #)
1	Programming Basics	1, 2
2	Variables and Operators	1, 2
3	Loops	1, 2, 3
4	Control Statements	1, 2, 3
5	Input/Output	1, 2, 3
6	Arrays and Structures	1, 2, 3
7	Functions	1, 2, 3
8	Recursion	1, 2, 3, 4
9	Strings	1, 2, 3, 4

10	Pointers	1, 2, 3, 6
11	Dynamic Memory Allocation	1, 2, 3, 6
12	Search on Arrays	1, 2, 3, 5
13	Sorting	1, 2, 3, 5
14	Review	1, 2, 3, 4, 5, 6

Learning and Teaching Methods

Course Learning Outcomes	Learning and Teaching Methods
1. Utilize modern software development tools and methods. Such skills will be useful in students' future computer science courses, as well as in their future careers.	Laboratories Lab Work; Programming Assignments; Quizzes; Exam
2. Have proficiency in writing and explaining small to mid-sized structured programs which will incorporate fundamental language elements	Laboratories Lab Work; Programming Assignments; Quizzes; Exam
3. Solve abstract computational problems through the use of programs.	Laboratories Lab Work; Programming Assignments; Quizzes; Exam
4. Write their own recursive functions and analyze other's recursive code for correctness.	Laboratories Lab Work; Programming Assignments; Quizzes; Exam
5. Identify and compare common algorithms used in searching and sorting, and learn how to implement them in C.	Laboratories Lab Work; Programming Assignments; Quizzes; Exam
6. Use pointers and memory management in the C language and be able to implement and explain code using these features.	Laboratories Lab Work; Programming Assignments; Quizzes; Exam

Laboratory Content

Starting the second week of class, we will conduct regular lab sessions in our computer lab, where students will be tasked with performing timed programming tasks which will be integrated into the larger programming assignments for the course.

E-Learning

This course was approved by SEDS Curriculum Committee in Fall 2014 to be run using the "Flipped Classroom" approach. The course has been internally assessed and subsequently refined on a regular basis via student evaluations and questionnaires, faculty comments, and analysis of student performance data.

Pre-recorded video lectures created for the course will be utilized to complement the real-time lectures.

While not an online course, some of the lectures may be delivered virtually according to

the published course schedule.

Grading

Undergraduate

Letter Grade	Percent Range	Grade (quality points)	Grade Descriptors
A	95-100	4.00	Student has exceeded expectations for all LOs, with evidence of highly comprehensive and in-depth knowledge, understanding and skills
A-	90-94.9	3.67	
B+	85-89.9	3.33	Student has achieved all the LOs of the assessment, exceeding expectations for some LOs, with evidence of thorough knowledge, understanding and skills.
B	80-84.9	3.00	
B-	75-79.9	2.67	
C+	70-74.9	2.33	Student has achieved all the LOs of the assessment with evidence of satisfactory knowledge, understanding and skills.
C	65-69.9	2.00	
C-	60-64.9	1.67	
D+	55-59.9	1.33	Student has partially achieved the LOs of the assessment with evidence of only limited knowledge, understanding and skills.
D	50-54.9	1.00	

F	0-49.9	0	Student has failed to achieve the LOs of the assessment with very little or no evidence of knowledge, understanding and skills.
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Key Books and Learning Materials

Software				
#	Title	Years and Edition	Publisher/ Manufacturer	Description
1	Eclipse, OBS, Zoom			

Academic Integrity Statement

[Student Code of Conduct and Disciplinary Procedures](#)

The specific guidelines are published in the NU Student Handbook. In particular,

- The assignments in this class are designed to introduce important concepts and techniques, and enable you to explore the material independently so as to gain insight and comprehension of the subject.
- The course is designed such that the new material presented each lesson builds on the skills developed in the preceding days; thus, any action that interferes with this process (e.g., skipping lesson exercises, copying) will seriously impede your progress.
- You are welcome—and encouraged—to talk through concepts and ideas with your fellow students and to study with them, but do not give or receive direct help from your classmates on graded exercises.
- Assignments should be completed individually. If you distribute or allow others to look at your work, even if you are not intending them to copy it, this is still considered academic misconduct.
- Even the appearance of cheating or inappropriate copying should be avoided.
- Using code that you find online is strictly forbidden, as is using code generation tools such as ChatGPT, MS Copilot, GitHub Copilot, etc.
- Students should be aware that the code submission process incorporates automated detectors for plagiarism and use of code generation tools.
- You may only get help on graded work from designated people—the instructors, TAs, or lecturers for the course. If you are struggling with something, by all means, please seek help from them.

In the event that academic misconduct such as plagiarism or cheating or use of code generation tools (AI) is discovered, the student will receive no credit for the work, and the event will be reported to the Dean of your school. Egregious cases, or a second offense,

can result in failure of the course and potential suspension or expulsion from the university.

All submissions will be electronically examined for indications of potential code plagiarism using the Stanford University Measure of Software Similarity (MOSS) tool; examples of similarity assessment can be seen on our course Moodle page. Submissions with high similarity scores will be subject to further review as possible instances of academic misconduct; in such cases, students will be notified (within 7 days) and provided the opportunity to address the concerns.

Course Expectations

The Nazarbayev University Special Learning Needs Committee (SLNC) is committed to creating an equitable and inclusive education environment for all students. If you have a qualified special learning need (physical, cognitive, socio-emotional, and psychological), please contact the SLNC as early as possible to ensure you receive the fullest support available.

For more information, please contact us at SLNC@nu.edu.kz and follow this [link](#) to the Accommodation Request Instructions.

Course Expectations

ATTENDANCE

Lab Work, Midterm and Final Exam components must be carried out and submitted in class within the given timeframe.

ELECTRONIC RESOURCES

Students will need to have their own laptops with the proper software installed.

Students will need to use an IDE (Integrated Development Environment) to develop, run, and debug our C programs, choosing among the ones that are available online for free download to their own computer.

Students are expected to check their Nazarbayev University e-mail on a daily basis for updates and announcements about the course. Not checking their e-mail is not an excuse for missing an announcement.

LAB SUBMISSION POLICY

Students will also be required to use Moodle to submit their lab exercises and programming assignments when directed. These need to be submitted at the time and date specified by their instructors. The lab assignments are meant to be individual.

In general, there is no late policy -- if you submit an assignment after it is due, you get zero points for your assignment. In cases of illness or family emergency, you must inform your instructor immediately if you believe you will not be able to submit your assignment on time, and a spravka needs to be submitted to the university for verification. In such cases, an exception may be made at the discretion of your instructor.

GRADE POLICY

The passing grade for the course is D and above. However, the course may serve as a prerequisite for another course that requires a higher minimum grade. For example, students must earn a grade of C- or better in CSCI 151 to enroll in CSCI 152. As per CS department policy, students with failing marks within 5% of the passing threshold do not have the option of receiving an Additional Assessment Task at the end of the semester to increase their grade.

Supporting Documents

Course Approval Date

School Learning and Teaching Committee Approval:

Course Origination Date: 2026-05-09

Effective from: Spring 2023